

PATIENT INFORMATION

Patient Name: Sarah Smith
Age / Sex: 34 / Female
Date of Birth: 05/14/1990
Medical Record #: SS-342199
Accession #: CS-24-15872
Specimen ID: 24-4581 A1
Specimen Type: Lung, Left Upper Lobe Biopsy
Collection Date: 04/29/2024
Received Date: 05/02/2024
Report Date: 05/13/2024

Ordering Physician: Alexandra Johnson, MD
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CLINICAL INFORMATION

Clinical Indication: Non-small cell lung carcinoma. Evaluate for clinically significant genomic alterations to aid in therapeutic decision making.

SUMMARY OF RESULTS

The tumor molecular profile is summarized below. Please see subsequent pages for complete results and interpretation.

BIOMARKER	METHOD	RESULT	CLINICAL INTERPRETATION
EGFR (SNVs/Indels)	NGS	Negative	No clinically significant EGFR alterations detected
KRAS	NGS	p.G12C Detected	Actionable mutation detected
ALK (Fusion)	RNA Seq	Negative	No ALK fusion detected
ROS1 (Fusion)	RNA Seq	Negative	No ROS1 fusion detected
RET (Fusion)	RNA Seq	Negative	No RET fusion detected
MET Exon 14 Skipping	RNA Seq	Negative	No MET exon 14 skipping detected
BRAF	NGS	Negative	No clinically significant BRAF alterations detected
PD-L1 (22C3) TPS	IHC	65%	High PD-L1 expression
Tumor Mutational Burden	WES	12 mut/Mb	High
Microsatellite Instability	NGS	Stable (MSS)	No evidence of MSI
Homologous Recombination Deficiency	NGS	Negative	HRD not detected

PATHOGENIC VARIANTS

GENE	VARIANT	EFFECT	VAF (%)	CLINICAL SIGNIFICANCE
KRAS	c.34G>T (p.G12C)	Missense	38%	Oncogenic
TP53	c.818G>A (p.R273H)	Missense	44%	Pathogenic
STK11	c.1060T>C (p.F354L)	Missense	29%	Pathogenic
KEAP1	c.1154G>A (p.G385D)	Missense	27%	Likely Pathogenic

GENES TESTED (Selected)

GENE	RESULT	GENE	RESULT
AKT1	Not Detected	NTRK2	Fusion Not Detected
BRCA1	Not Detected	NTRK3	Fusion Not Detected
BRCA2	Not Detected	RET	Fusion Not Detected
CDKN2A	Not Detected	BRAF	Not Detected
ERBB2 (HER2)	Not Detected	MET Amplification	Not Detected
PIK3CA	Not Detected	CCND1 Amplification	Not Detected
NRAS	Not Detected	MYC Amplification	Not Detected
PTEN	Not Detected	CDK4 Amplification	Not Detected
MAP2K1	Not Detected	MDM2 Amplification	Not Detected
FGFR1	Not Detected	EGFR Amplification	Not Detected
FGFR2	Not Detected	ALK Amplification	Not Detected
FGFR3	Not Detected	ROS1 Amplification	Not Detected
IDH1	Not Detected	PDGFRA	Not Detected
IDH2	Not Detected	KIT	Not Detected
NTRK1	Fusion Not Detected	FLT3	Not Detected

GENOMIC SIGNATURES

SIGNATURE	RESULT	INTERPRETATION
Microsatellite Instability (MSI)	Stable (MSS)	No evidence of MSI
Tumor Mutational Burden (TMB)	12 mut/Mb	High (≥10 mut/Mb)
Homologous Recombination Deficiency (HRD)	Negative	HRD not detected
Loss of Heterozygosity (LOH)	Low	Low genomic instability
Cancer Type Relevant Pathways	3 Activated	See report

METHODS

Next-generation sequencing (NGS) was performed on tumor DNA and RNA using targeted capture and whole transcriptome sequencing. Immunohistochemistry (IHC) was performed on formalin-fixed, paraffin-embedded tumor tissue using FDA-approved assays.

DISCLAIMER

This test was developed and its performance characteristics determined by Caris Life Sciences. It has not been cleared or approved by the U.S. Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. This test is used for clinical purposes. It should not be regarded as investigational or for research.



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Medical Director